

LESSON 3-5 • SESSION 2 • PRACTICE • FORM B

Comparing Rates with Benchmarks

Learning target – I can use a benchmark to compare two rates without dividing. • 6.RP.A.3b

REMEMBER

- A **benchmark** is one amount you use for **both** choices. Scale each rate to it, then compare.
- Pick a benchmark both amounts divide into evenly: 12 works for 4 and 6.
- You can also **divide** to find each **unit rate** and compare. Round money to the nearest cent.

WORKED EXAMPLE

Hose A fills 6 gallons in 2 minutes. Hose B fills 10 gallons in 4 minutes. Which is faster?

Step 1 Benchmark: 20 minutes.

Step 2 A: $6 \times 10 = 60$ gallons. B: $10 \times 5 = 50$ gallons.

Step 3 Check: $6 \div 2 = 3$ and $10 \div 4 = 2.5$ gallons per minute.

Answer Hose A is faster

WORDS benchmark (*punto de referencia*) • rate (*tasa*) • unit rate (*tasa unitaria*)

PRACTICE

show your work • label units

1. Find each unit rate.

a. 75 flyers in 5 min ____ per min

b. 96 stairs in 8 min ____ per min

c. \$18 for 8 lb ____ per lb

d. 320 miles on 10 gal ____ per gal

2. **Sticker Machines.** Machine A costs 2 coins and gives 5 stickers. Machine B costs 3 coins and gives 8 stickers. Use a 30-coin benchmark.

Hint Machine A $\times 15$; Machine B $\times 10$.

	Coins	Stickers	Stickers for 30 coins
Machine A	2	5	
Machine B	3	8	

Better machine _____

3. **Typing Contest.** Tyler types 156 words in 4 minutes. Sarah types 190 words in 5 minutes. Who types faster?

Hint Find words per minute for each.

Show your work.

Faster _____

4. **Gummy Bears.** Store A sells 4 lb for \$9. Store B sells 6 lb for \$14. Use a 12-lb benchmark to find the better buy.

Show your work.

Better buy _____

5. **Corner Store Chips.** Store J sells 4 bags of chips for \$10.00. Store K sells 6 bags for \$14.40.

a. Price per bag: Store J ____ Store K ____

b. Cost of 12 bags: Store J ____ Store K ____

c. Which store is the better buy? _____

6. **Ice Rink Passes.** Rink A: 4 sessions for \$34.00. Rink B: 6 sessions for \$52.20. Find the cost per session. Which is the better deal?

Hint Divide each price by the number of sessions.

Show your work.

Better deal _____

7. **Reading Rates.** Reader 1 finishes 35 pages in 5 days. Reader 2 finishes 48 pages in 6 days.

Hint $30 \div 5 = 6$ and $30 \div 6 = 5$.

- a. Pages in 30 days: Reader 1 _____ Reader 2 _____
- b. Who reads faster? _____

Show your work.

8. **Field Trip Vans.** Van company A charges \$120 for 8 students. Company B charges \$165 for 11 students.

- a. Use 88 students as a benchmark. Find each total. _____
- b. Which company is cheaper per student? _____
- c. Why might a benchmark be easier than dividing? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Maya’s Work.** Ride A costs 3 tickets for 12 points. Ride B costs 5 tickets for 15 points. Maya subtracted $12 - 3 = 9$ and $15 - 5 = 10$, so she picked Ride B. Explain her mistake and find the better ride.

TEST PRACTICE

choose the best answer

10. A good benchmark for comparing a 5-pack and a 3-pack is –
- A** 8 **B** 10 **C** 15 **D** 20
11. A benchmark amount must be one that both rates can –
- A** reach exactly **B** add to **C** subtract from **D** round to

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

\$28.80 G	\$27 R	15 B	\$17 Z	240 I	288 X	\$0.12 Q	\$8.50 A	\$1,320 N	80 A
#1a		#2	#4	#5b	#6	#7a	#8a		

CHECK YOURSELF

☐ I can do this on my own
☐ I need a little help
☐ I have a question about #____